

MATH 220/199 Week 14 Day 2

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

Does the Over or Under Hit?

In order to find areas between curves, we need to find which function is the top, and which is the bottom. The following exercises help you with some ideas to find which is which without graphing.

1. Prove the following statements WITHOUT graphing: Let $f(x) = x, g(x) = \sin(x)$

- (a) For all $x \in [-\frac{\pi}{2}, 0), g(x) > f(x)$.
- (b) For all $x \in (0, \frac{\pi}{2}], f(x) > g(x)$.

2. Over the interval $(-\pi, \pi)$, determine the intervals where $\cot(x) > \cos(x)$.

3. On which interval(s) is $\sqrt{|x|} > |x|$?

General Practice

1. (FTC) Compute $\frac{d}{dx} \int_3^{\sin^2(x)} \frac{t}{1+t^2} dt$

2. (Riemann Sums) For $f(x) = x^2 - 1$ over the interval $[0, 8]$, compute each of the following. For the first two, state whether they are and overestimate or underestimate.

(a) R_4

(b) L_4

(c) M_4

3. (U-Sub) Compute $\int (1 - \frac{1}{x}) \cos(x - \ln x) dx$